

Neil Motiani

(917) 330-4595 | neilmotiani@gmail.com | [linkedin/neil-motiani](https://www.linkedin.com/in/neil-motiani) | [github/neilmotiani](https://github.com/neilmotiani)

EDUCATION

Stevens Institute of Technology

B.S. in Computer Science – Dean's List

Hoboken, NJ

Graduating Dec. 2027

Coursework: Machine Learning, Data Structures, Algorithms, Linear Algebra, Computer Architecture and Organization, Discrete Mathematics, Probability and Statistics

Activities: Software Engineering Club, Python for Business, Stevens Society of AI, SASA, Flag Football

EXPERIENCE

CapitalBooks

2026 – Present

Co-Founder

Remote

- Engineered a conversational AI agent that autonomously selects and orchestrates 23 custom tools in parallel, eliminating manual routing logic and enabling reliable execution of complex, multi-step user requests end-to-end, from initial intent recognition through tool selection, execution, and final response synthesis back to the end user.
- Cut end-to-end response latency from 35s to under 5s (86% reduction) by redesigning the request pipeline with async parallelism, response caching, and intelligent model routing, substantially boosting product responsiveness, perceived reliability, and user retention across high-traffic conversational sessions and peak usage periods.
- Built a self-improving classification system that learns from human corrections via a continuous feedback loop, cutting AI model dependency and inference costs over time while maintaining classification accuracy across evolving transaction types, edge cases, and shifting user behavior patterns as the product scaled to more users.

Tumbles

Aug. 2025 – Dec. 2025

Data Analytics Intern

Remote

- Eliminated 15+ hours of weekly manual reporting by building automated pipelines and dashboards in Python and SQL over Google Ads, Meta Ads, Google Analytics, and CRM data, improving funnel visibility for the marketing, operations, and leadership teams overseeing enrollment strategy across every campaign.
- Reduced cost-per-acquisition by 15% by building and evaluating propensity-to-enroll, cohort retention, and revenue forecasting models in pandas and scikit-learn, then running A/B and uplift tests across channels to guide data-driven budget reallocation decisions on a recurring weekly and monthly basis.
- Designed scalable data schemas, defined KPIs, and established reporting workflows linking multi-channel traffic to downstream revenue and consumer behavior, enabling data-driven decisions on channel-mix optimization and long-term enrollment growth across the entire organization and all business units.

PROJECTS

Macro Regime Radar | *Python, pandas, yfinance, scikit-learn, Streamlit, GitHub Actions*

Aug. 2026

- Built and automated a daily-refreshing data pipeline ingesting 20+ years of gold, crude oil, US Dollar Index, 10-year Treasury yield, and VIX data (5,700+ observations), resolving cross-asset calendar mismatches and a negative-price futures-expiry anomaly, then engineered rolling correlation, volatility, and momentum features across five asset pairs to quantify shifting macro relationships.
- Applied unsupervised learning (k-means, Gaussian Mixture Models) to classify 20+ years of market history into five interpretable macro regimes (calm, dollar-driven, safe-haven, commodity-macro, crisis), cross-validating the two independent methods to confirm 209 of 211 crisis-period days matched (Adjusted Rand Index 0.39) – correctly and independently identifying the 2008 and 2020 crises with no labeled data.
- Deployed an interactive Streamlit dashboard with regime-shaded price charts, live confidence scoring, and curated macro-event annotations, backed by a GitHub Actions pipeline that scores new trading days against a frozen, versioned model daily – avoiding retroactive relabeling of historical regimes while keeping the dataset continuously current.

Prediction-Market Arbitrage Engine | *Python, FastAPI, asyncio, PostgreSQL, Next.js, Docker*

Sept. 2026

- Built an asynchronous prediction-market arbitrage engine that ingests real-time Polymarket order books, discovers and validates binary contracts, and detects complement and cross-venue pricing inefficiencies, evaluating depth-aware execution costs including fees, slippage, liquidity, latency, and capital constraints before flagging any trade.
- Developed a FastAPI, PostgreSQL, and Next.js trading research platform with WebSocket updates and a dark-themed quantitative dashboard, implementing automatic paper trading with persistent simulated positions, duplicate-fill protection, risk limits, and venue-confirmed settlement tracking end-to-end.
- Processed 3,432 real-market snapshots and 1,716 arbitrage checks across 12 markets in a live observation window, rejecting opportunities that failed execution constraints; validated the system with 59 automated tests and benchmarked mock-path performance at 7,230 snapshots/sec and 4,820 checks/sec with 0.105ms p95 detection latency.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java (Introductory), OCaml (Introductory)

Libraries & Frameworks: scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, Streamlit, TensorFlow

Developer Tools: Git/GitHub, Docker (Introductory), Jupyter Notebook, VS Code, Tableau, Google Colab, Slack